

Assignment – 20:

Problem Statement:

Write a C function to swap two numbers with and without using pointers.

Code:

```
#include <stdio.h>

void swap_simple(int a,int b)
{
    int temp;
    temp=a;
    a=b;
    b=temp;
    printf("After swapping without pointers:\n");
    printf("a = %d, b = %d\n",a,b);
}

void swap_pointers(int *a,int *b)
{
    int temp;
    temp=*a;
    *a=*b;
    *b=temp;
    printf("After swapping with pointers:\n");
    printf("a = %d, b = %d\n",*a,*b);
}

int main()
{
    int x,y;
    printf("Enter numbers for swapping: ");
    scanf("%d%d",&x,&y);
    swap_simple(x,y);
```

```
    swap_pointers(&x,&y);  
}
```

Input & Output:

```
Enter numbers for swapping: 12 13  
After swapping without pointers:  
a = 13, b = 12  
After swapping with pointers:  
a = 13, b = 12
```

```
Enter numbers for swapping: -123 907  
After swapping without pointers:  
a = 907, b = -123  
After swapping with pointers:  
a = 907, b = -123
```

